

Mohamed Abdelrahman

832-419-9579 | msabdelrahman02@gmail.com | LinkedIn | GitHub | Portfolio

EDUCATION

University of Houston

Bachelor of Science in Computer Science - Major GPA: 3.81

Expected Graduation: Dec 2026

Houston, TX

EXPERIENCE

Site Reliability Engineer, Intern

IBM

May 2026 – Present

Austin, TX (Hybrid)

- Built a rule-based label migration tool in Python with an LLM fallback for ambiguous cases to resolve mislabeled fields across 4,000+ SRE Jira issues, making thousands of hours in team credit trackable
- Built a direct API client with token bucket-style request dispatch after the team's single-request LLM interface became a bulk-processing bottleneck, tuning it to 128 concurrent workers that could run continuously without overloading the API
- Built a schema-driven, multi-prompt LLM pipeline to analyze 243 SRE incident reports in parallel, extracting scope, timeline, root cause, customer impact, and RCA per issue, cutting analysis time from 2.8 hours to ~10 minutes (17x speedup)
- Automated closure of 250+ stale SRE Jira issues with a rule-based pipeline analyzing status and activity history, reducing backlog noise for the on-call team
- Built a configurable, diff-based label migration tool with dry-run and rollback support for safe production rollout of label changes

Course Facilitator

Department of Computer Science - University of Houston

Jan 2026 – Present

Houston, TX

- Prepare coding problems and instructional material to facilitate COSC 3320: Algorithms and Data Structures
- Host two workshops per week, coordinating with faculty and facilitators to ensure effective learning experiences

Software Engineer (Consultant)

bld.ai

Nov 2022 – Apr 2023

Remote

- Developed a full-stack platform using React and Django to match families for playdate arrangements
- Refactored database models, authorization, and data validation, reducing bugs in the codebase by 50%
- Deployed a minimum viable product (MVP) within one month using a two-week sprint cycle
- Designed and configured role-based permission groups to control access while ensuring data integrity

LEADERSHIP

Director of Coding Competitions

IEEE-NSM - University of Houston

Jan 2025 – Present

- Prepared and hosted workshops on data structures and algorithms in C++, and curated competitive programming competitions to strengthen undergraduate students' algorithmic problem-solving skills
- Organized practice sessions for the ICPC team by selecting problems and arranging virtual contest sessions, contributing to the team's 1st place finish at UH Site
- Coordinated with students and faculty on ICPC hosting logistics at UH, organizing scheduling shifts for 20 volunteers to ensure successful contest execution

AWARDS

Golden Medalist, ICPC South Regional Contest - UH Site

University of Houston

Nov 2025

- Secured 1st place at UH Site (16th place regionally) in the Division 1 ICPC South Regional Contest by solving 4 problems using C++ and advanced algorithmic techniques

PROJECTS

TallyCart | *Django, PostgreSQL, Celery/Redis, Docker, LLM Integration*

- Architected a Django/PostgreSQL backend with Celery/Redis background processing, isolating OCR, LLM, and pricing logic behind swappable interfaces across 7 modules with 30+ tests
- Built a receipt-to-data pipeline chaining OCR with LLM-based extraction across three interchangeable providers with automatic failover, secured via custom JWT authentication

TECHNICAL SKILLS

Languages: C++, Python, SQL, JavaScript, HTML/CSS, Dart, C#

Frameworks: Django, React, NextJS, Flutter, Unity

Tools: Git, Docker, Postman, Coverage, GitHub Actions, VS Code, Ubuntu Linux, Bash